Kerala Polytechnic Revision 2021 • Semester 1 • Basic Science

Mathematics I

Complete beginner-friendly digital textbook for Course 1002. It works as textbook, revision guide, workbook, formula book and exam preparation guide in one offline HTML file.

Designed for first-semester diploma students who may be weak in mathematics. Learn the idea, copy the method, practise the example, then check the answer key.

Course Details

1002 Course Code	**5** Credits
75 Total Periods	**5-0-0** L:T:P
Basic Science Category	**Theory** Subject Type

Main areas: Complex Numbers, Coordinate Geometry, Trigonometry, Limits, Differentiation I and Differentiation II.

Course Overview

What this subject teaches

Why it matters

Complex numbers are used in electrical and electronics calculations. Straight lines help in graph interpretation. Trigonometry is used in waves, AC circuits, surveying and mechanics. Differentiation explains rate of change and slope.

How to study

1. Memorise formula with meaning of symbols.
2. Write one solved example by hand.
3. Try the matching practice question.

4. Revise mistakes before series tests.

Objective

To provide introductory coverage of Trigonometry, Differential Calculus and basic elements of Algebra for diploma engineering students.

Prerequisite

Secondary school algebra, factorisation, equations, coordinate plane, angle measurement, basic trigonometric ratios and simple graphs.

Course outcomes and hours

CO	Outcome	Hours	Level
CO1	Use complex numbers and equations of straight lines.	20	Applying
CO2	Solve problems related to trigonometry.	18	Applying
CO3	Use limits and derivatives to solve problems.	20	Applying
CO4	Apply composite, parametric, implicit and successive differentiation.	15	Applying

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Complete textbook map

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All important formulas

Worked Examples

38 solved examples

Course 1002

Practice Questions

Easy, medium, exam level

Quick Revision

Last-minute guide

Answer Key

Final answers and hints

Module 1 • CO1 • 20 hours

Complex Numbers and Straight Lines

Learn numbers of the form a+ib, then learn how straight lines are written, compared and solved in coordinate geometry.

Complex numbers

Definition, real part, imaginary part, i and $i^2=-1$ close

A complex number is $z=a+ib$. Here a is real part and b is imaginary part. The symbol i is defined by $i^2=-1$.

Formula: $Re(z)=a, Im(z)=b$

Basic: For 5-3i, Re=5 and Im=-3. **Exam:** If (2x+1)+i(y-4)=7+3i, then x=3, y=7.

Mistake: imaginary part of 5-3i is -3, not -3i.

Try: Find Re and Im of -2+9i.

Equality, addition and subtraction close

Equal complex numbers have equal real parts and equal imaginary coefficients. Add or subtract real parts together and imaginary parts together.

$$(a+ib)+(c+id)=(a+c)+i(b+d)$$
$$(a+ib)-(c+id)=(a-c)+i(b-d)$$

Basic: $(3+2i)+(5-7i)=8-5i$. **Exam:** $(8-3i)-(2+6i)=6-9i$.

Mistake: Do not mix real part with imaginary coefficient.

Try: $(4+9i)-(7-2i)$.

Multiplication, conjugate, modulus, amplitude, Cartesian and polar idea close

Multiply using brackets and replace i^2 by -1. Conjugate changes sign of imaginary part. Modulus is distance from origin. Amplitude is angle idea. Polar conversion is excluded, so know the idea only.

$$(a+ib)(c+id)=(ac-bd)+i(ad+bc)$$
$$\bar{z}=a-ib,\ |z|=\sqrt{(a^2+b^2)},\ \tan\theta=b/a$$
$$\text{Cartesian: } a+ib;\ \text{Polar idea: } r(\cos\theta+i\sin\theta)$$

Basic: $(2+3i)(4+i)=5+14i$. **Exam:** For $z=3+4i$, $\bar{z}=3-4i$ and $|z|=5$.

Mistake: i^2 is -1. Modulus is never negative.

Try: Find conjugate and modulus of $8-15i$.

Straight lines

Forms of straight line close

A straight line equation may be selected depending on the available data: slope, one point, two points, intercepts or general equation.

$$m=(y_2-y_1)/(x_2-x_1)$$
$$y=mx+c;\ y-y_1=m(x-x_1)$$
$$(y-y_1)/(y_2-y_1)=(x-x_1)/(x_2-x_1)$$
$$x/a+y/b=1;\ ax+by+c=0$$

Basic: slope through (1,2),(3,6) is 2. **Exam:** slope 3 through (2,5): y-5=3(x-2), so y=3x-1.

Mistake: Reversing only one difference changes sign.

Try: line with slope -2 through (1,4).

Intersection, angle, parallel, perpendicular and distance close

Intersection satisfies both equations. Angle between lines uses their slopes. Parallel lines have equal slopes. Perpendicular lines have product -1. Distance formula gives shortest distance from point to line.

$$\tan\theta=|(m_2-m_1)/(1+m_1m_2)|$$
$$\text{Parallel: } m_1=m_2;\ \text{Perpendicular: } m_1m_2=-1$$
$$\text{Distance}=|ax_1+by_1+c|/\sqrt{(a^2+b^2)}$$

Basic: x+y=5 and x-y=1 gives intersection (3,2). **Exam:** Distance from (2,-1) to 5x-12y+6=0 is 28/13.

Mistake: Use modulus in distance formula.

Try: Are slopes 3 and -1/3 perpendicular?

Module 2 • CO2 • 18 hours

Trigonometry

Trigonometry connects angles and ratios. Focus on formulas, standard values, signs and simple applications.

Angles

Degree divides a revolution into 360 parts. Radian is the angle made when arc length equals radius.

$180° = \pi$ rad

Degree to radian: **$\theta\pi/180$**

Radian to degree: **$\theta \times 180/\pi$**

Calculator mode alert: degrees for degree problems, radians for limits.

Ratios

$\sin\theta$=opposite/hypotenuse

$\cos\theta$=adjacent/hypotenuse

$\tan\theta$=opposite/adjacent

cosec=1/sin, sec=1/cos, cot=1/tan

Standard values

θ	0°	30°	45°	60°	90°
sinθ	0	1/2	1/√2	√3/2	1
cosθ	1	√3/2	1/√2	1/2	0
tanθ	0	1/√3	1	√3	Not defined

Identities, allied and formula groups close

Group	Formula	Use
Identities	$\sin^2A+\cos^2A=1$; $1+\tan^2A=\sec^2A$; $1+\cot^2A=\mathrm{cosec}^2A$	Simplification
Allied	$\sin(90°-A)=\cos A$; $\cos(90°-A)=\sin A$	Complement angles
Compound	$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B$; $\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$; $\tan(A\pm B)=(\tan A\pm\tan B)/(1\mp\tan A\tan B)$	Exact values
2A and 3A	$\sin2A=2\sin A\cos A$; $\cos2A=\cos^2A-\sin^2A$; $\tan2A=2\tan A/(1-\tan^2A)$; $\sin3A=3\sin A-4\sin^3A$; $\cos3A=4\cos^3A-3\cos A$	Multiple angles
Product and sum	$2\sin A\cos B=\sin(A+B)+\sin(A-B)$; $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	Transformation

Sign memory: ASTC. I quadrant all positive, II sine, III tan, IV cos.

Limits and Differentiation I

A function is an input-output rule. A limit is the value approached. A derivative is rate of change or slope.

Function and limits

If $f(x)=x^2+1$, then $f(3)=10$. For limits, first try direct substitution. If 0/0 comes, simplify.

$$\lim_{x\to a} f(x)$$
$$\lim_{x\to 0} \sin x/x=1$$
$$\lim_{x\to 0} \tan x/x=1$$

Differentiation by definition

$$f'(x)=\lim_{h\to 0}[f(x+h)-f(x)]/h$$

After understanding the definition, use standard derivative rules for simple problems.

Standard derivatives

y	dy/dx	Care
x^n	nx^{n-1}	Power rule
sinx	cosx	Radian idea
cosx	-sinx	Negative sign
tanx	$\sec^2 x$	Standard
e^x	e^x	Same
logx	1/x	Natural log
$\sin^{-1}x$	$1/\sqrt{(1-x^2)}$	Inverse trig
$\cos^{-1}x$	$-1/\sqrt{(1-x^2)}$	Negative
$\tan^{-1}x$	$1/(1+x^2)$	Inverse trig

Differentiation rules close

$d(u \pm v)/dx = du/dx \pm dv/dx$
$d(ku)/dx = k\,du/dx$
$d(uv)/dx = u\,dv/dx + v\,du/dx$
$d(u/v)/dx = (v\,du/dx - u\,dv/dx)/v^2$

Common errors: wrong substitution in limits, missing brackets, derivative of cosx without minus sign, wrong product rule.

Module 4 • CO4 • 15 hours

Differentiation II

Use chain rule for function of a function, parametric derivative for x and y in t, implicit differentiation for mixed x-y equations and successive differentiation up to second order.

Chain rule

If $y=f(u)$, $u=g(x)$, $dy/dx = dy/du \times du/dx$

Flow: outer derivative × inner derivative. Avoid highly nested functions excluded in syllabus.

Parametric

$dy/dx = (dy/dt)/(dx/dt)$

Differentiate y and x separately with respect to t, then divide.

Implicit

$d(y^2)/dx = 2y\,dy/dx$

When x and y are mixed, differentiate both sides and collect dy/dx terms.

Second derivative

dy/dx = first derivative; d^2y/dx^2 = second derivative

Successive differentiation table

y	dy/dx	d²y/dx²
x^4	$4x^3$	$12x^2$
sinx	cosx	-sinx
cosx	-sinx	-cosx
e^x	e^x	e^x

Formula Bank

All formulas in one place

Area	Formula	Where used	Mini example	Mistake
Complex	z=a+ib	Identify parts	3+4i	Im is 4, not 4i
Conjugate	z̄=a-ib	Change sign	2-5i⇒2+5i	Changing real sign
Modulus	$\lvert z\rvert=\sqrt{a^2+b^2}$	Magnitude	$\lvert 3+4i\rvert=5$	Negative answer
Slope	$m=(y_2-y_1)/(x_2-x_1)$	Line steepness	m=2	Wrong order
Line	$y=mx+c$; $y-y_1=m(x-x_1)$	Equation of line	y=2x+1	Wrong c
Distance	$\lvert ax_1+by_1+c\rvert/\sqrt{a^2+b^2}$	Point to line	3x+4y-5	No modulus
Conversion	180°=π rad	Angle units	60°=π/3	Mode error
Identity	$\sin^2A+\cos^2A=1$	Simplify	$1-\cos^2A=\sin^2A$	No square
Compound	sin(A+B)=sinAcosB+cosAsinB	Exact value	sin75°	Sign
Double	sin2A=2sinAcosA	2A problems	A=30°	Wrong formula
Limit	lim sinx/x=1	Trig limits	$x \to 0$	Use radians
Power	$d(x^n)/dx=nx^{n-1}$	Derivative	$x^5 \Rightarrow 5x^4$	nx^n
Product	d(uv)=u dv+v du	Product	xsinx	Multiplying derivatives
Quotient	$d(u/v)=(vdu-udv)/v^2$	Fraction	x/(x+1)	Wrong order
Chain	dy/dx=dy/du×du/dx	Composite	$(3x+2)^4$	Missing inner
Parametric	dy/dx=(dy/dt)/(dx/dt)	x,y in t	$x=t^2, y=t^3$	Reverse division
Second	d^2y/dx^2	Differentiate twice	$x^3 \Rightarrow 6x$	Stop early

Solved Examples Bank

38 solved examples

M1-E1 Add

3+4i and 5-7i.

Real 3+5=8, imaginary 4-7=-3.

8-3i

M1-E2 Subtract

(6-2i)-(1+9i).

6-1=5, -2-9=-11.

5-11i

M1-E3 Multiply

(2+3i)(4-i).

$8-2i+12i-3i^2=11+10i$.

11+10i

M1-E4 Conjugate

z=7-24i.

$\bar{z}=7+24i$, $|z|=\sqrt{625}$.

$\bar{z}=7+24i$, $|z|=25$

M1-E5 Equality

x+2i=5+yi.

Equate real and imaginary parts.

x=5,y=2

M1-E6 Slope

(2,3),(6,11).

m=(11-3)/(6-2).

m=2

M1-E7 Line

Slope 4 through (1,2).

y-2=4(x-1).

y=4x-2

M1-E8 Two-point line

(1,1),(3,5).

m=2, y-1=2(x-1).

y=2x-1

M1-E9 Intersection

x+y=7, x-y=1.

Add equations: 2x=8.

(4,3)

M1-E10 Distance

(2,1) to 3x+4y-10=0.

|6+4-10|/5.

0

M2-E1 Degree to radian

150°.

150π/180.

5π/6

M2-E2 Radian to degree

2π/3.

(2π/3)×180/π.

120°

M2-E3 Standard value

sin60°+cos60°.

$\sqrt{3}/2+1/2$.

$(\sqrt{3}+1)/2$

M2-E4 Identity

$1-\cos^2 A$.

Use $\sin^2 A+\cos^2 A=1$.

$\sin^2 A$

M2-E5 Allied

sin(90°-A).

Complement formula.

cosA

M2-E6 Compound

$\sin 75°$.

$\sin(45+30)=\sin 45\cos 30+\cos 45\sin 30$.

$(\sqrt{6}+\sqrt{2})/4$

M2-E7 Difference

$\cos 15°$.

$\cos(45-30)=\cos 45\cos 30+\sin 45\sin 30$.

$(\sqrt{6}+\sqrt{2})/4$

M2-E8 Double angle

$\sin A=3/5$, $\cos A=4/5$.

$\sin 2A=2\sin A\cos A$.

24/25

M2-E9 Triple angle

$\sin 3A$.

Use standard formula.

$3\sin A-4\sin^3 A$

M2-E10 Product

$2\sin A\cos B$.

Product-to-sum formula.

$\sin(A+B)+\sin(A-B)$

M3-E1 Function

$f(x)=x^2+2x$, $f(3)$.

9+6.

15

M3-E2 Direct limit

$\lim_{x\to 2} (x^2+3)$.

Substitute 2.

7

M3-E3 Algebraic limit

$\lim_{x\to 1} (x^2-1)/(x-1)$.

Factor to $x+1$.

2

M3-E4 Trig limit

$\lim_{x\to 0} \sin x/x$.

Standard limit.

1

M3-E5 Power

$d(x^5)/dx$.

Power rule.

$5x^4$

M3-E6 Polynomial

$d(x^3+4x^2-7)/dx$.

Differentiate termwise.

$3x^2+8x$

M3-E7 Trig derivative

$d(\sin x+\cos x)/dx$.

$\cos x-\sin x$.

$\cos x-\sin x$

M3-E8 Product

$d(x\sin x)/dx$.

$x\cos x+\sin x$.

$x\cos x+\sin x$

M3-E9 Quotient

$d[x/(x+1)]/dx$.

$[(x+1)-x]/(x+1)^2$.

$1/(x+1)^2$

M3-E10 Exp-log

$d(e^x+\log x)/dx$.

Use standard derivatives.

e^x+1/x

M4-E1 Chain

$d(3x+2)^4/dx$.

$4(3x+2)^3 \times 3$.

$12(3x+2)^3$

M4-E2 Chain trig

d sin(5x)/dx.

cos(5x)×5.

5cos(5x)

M4-E3 Parametric

$x=t^2, y=t^3$.

$dy/dx=(3t^2)/(2t)$.

3t/2

M4-E4 Parametric

$x=2t+1, y=t^2$.

dy/dx=2t/2.

t

M4-E5 Implicit

$x^2+y^2=25$.

2x+2y dy/dx=0.

dy/dx=-x/y

M4-E6 Implicit

xy=10.

x dy/dx+y=0.

dy/dx=-y/x

M4-E7 Second derivative

$y=x^4$.

$y'=4x^3$, $y''=12x^2$.

$12x^2$

M4-E8 Second derivative

y=sinx.

y′=cosx, y″=-sinx.

-sinx

Practice Questions

Workbook practice

Easy

Module 1

1. Find Re and Im of 8-11i.
2. Add 7+2i and 3-9i.
3. Find conjugate of -4+6i.
4. Find slope through (0,2) and (4,10).

Medium

Module 1

1. Multiply (3+2i)(5-i).
2. Find line through (2,3) with slope -1.
3. Find intersection of x+y=9 and x-y=3.
4. Find distance from (1,1) to 3x+4y-12=0.

Exam Level

Module 2

1. Convert 225° into radians.
2. Find exact value of cos75°.
3. Simplify $(1-\sin^2 A)/\cos A$.
4. Transform 2cosAcosB.

Medium

Module 3

1. Evaluate $\lim_{x\to 2} (x^2-4)/(x-2)$.

Exam Level

Module 4

1. Differentiate $(2x-1)^5$.

Mixed

Quick Check

1. State product rule.

2. Differentiate x^4-5x^2+6.
3. Differentiate xcosx.
4. Differentiate $(x^2+1)/x$.

2. If $x=t^2+1$ and $y=t^3$, find dy/dx.
3. Differentiate $x^2+xy+y^2=7$ implicitly.
4. Find second derivative of $y=\cos x+x^3$.

2. State quotient rule.
3. Write sin2A formula.
4. Condition for perpendicular lines?

Quick Revision

Last-minute revision

Must know

- $i^2=-1$, $|a+ib|=\sqrt{(a^2+b^2)}$.
- Slope, line forms, angle and distance formulas.
- 180°=π rad and standard angle table.
- Direct substitution, factorisation and standard trig limits.
- Power, product, quotient and chain rules.

Common mistakes

- Wrong sign in cos derivative.
- Using degree mode for calculus limits.
- Forgetting inner derivative in chain rule.
- Not using dy/dx for y terms in implicit differentiation.
- Stopping at first derivative when second derivative is asked.

Master Answer Key

Answers for practice questions

M1 Easy: 1) Re=8 Im=-11. 2) 10-7i. 3) -4-6i. 4) m=2.

M1 Medium: 1) 17+7i. 2) y=-x+5. 3) (6,3). 4) distance=1.

M2: 1) 5π/4. 2) $(\sqrt{6}-\sqrt{2})/4$. 3) cosA. 4) cos(A+B)+cos(A-B).

M3: 1) 4. 2) $4x^3-10x$. 3) cosx-xsinx. 4) $1-1/x^2$.

M4: 1) $10(2x-1)^4$. 2) 3t/2. 3) dy/dx=-(2x+y)/(x+2y). 4) -cosx+6x.

Mixed: Product rule $d(uv)=u\,dv+v\,du$. Quotient rule $d(u/v)=(v\,du-u\,dv)/v^2$. $\sin 2A=2\sin A\cos A$. Perpendicular condition $m_1 m_2=-1$.

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